

A combination of prebiotic short- and long-chain inulin-type fructans enhances calcium absorption and bone mineralization in young adolescents.

BACKGROUND: Short-term studies in adolescents have generally shown an enhancement of calcium absorption by inulin-type fructans (prebiotics). Results have been inconsistent; however, and no studies have been conducted to determine whether this effect persists with long-term use. **OBJECTIVE:** The objective was to assess the effects on calcium absorption and bone mineral accretion after 8 wk and 1 y of supplementation with an inulin-type fructan. **DESIGN:** Pubertal adolescents were randomly assigned to receive 8 g/d of a mixed short and long degree of polymerization inulin-type fructan product (fructan group) or maltodextrin placebo (control group). Bone mineral content and bone mineral density were measured before randomization and after 1 y. Calcium absorption was measured with the use of stable isotopes at baseline and 8 wk and 1 y after supplementation. Polymorphisms of the Fok1 vitamin D receptor gene were determined. **RESULTS:** Calcium absorption was significantly greater in the fructan group than in the control group at 8 wk (difference: 8.5 +/- 1.6%; $P < 0.001$) and at 1 y (difference: 5.9 +/- 2.8%; $P = 0.04$). An interaction with Fok1 genotype was present such that subjects with an ff genotype had the least initial response to fructan. After 1 y, the fructan group had a greater increment in both whole-body bone mineral content (difference: 35 +/- 16 g; $P = 0.03$) and whole-body bone mineral density (difference: 0.015 +/- 0.004 g/cm²; $P = 0.01$) than did the control group. **CONCLUSION:** Daily consumption of a combination of prebiotic short- and long-chain inulin-type fructans significantly increases calcium absorption and enhances bone mineralization during pubertal growth. Effects of dietary factors on calcium absorption may be modulated by genetic factors, including specific vitamin D receptor gene polymorphisms.